

Wave 4.4 Latent-State Hysteresis Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `wave 4.4` causal latent-state / hysteresis-aware campaign; all six planned runs completed, and the program-level scalar winner did not change.

Campaign Artifacts

Artifact	Path
Campaign output	<code>output/training_campaigns/2026-06-16-18-06-11_track2h_latent_state_hysteresis_campaign_2026_06_16</code>
Leaderboard	<code>output/training_campaigns/2026-06-16-18-06-11_track2h_latent_state_hysteresis_campaign_2026_06_16/campaign_leaderboard.yaml</code>
Best-run pointer	<code>output/training_campaigns/2026-06-16-18-06-11_track2h_latent_state_hysteresis_campaign_2026_06_16/campaign_best_run.yaml</code>
Execution report	<code>output/training_campaigns/2026-06-16-18-06-11_track2h_latent_state_hysteresis_campaign_2026_06_16/campaign_execution_report.md</code>
Planning report	<code>doc/reports/campaign_plans/track_2/2026-06-16-16-00-57_track2h_latent_state_hysteresis_campaign_plan_report.md</code>
Campaign package	<code>config/training/track2h_latent_state_hysteresis/campaigns/2026-06-16_track2h_latent_state_hysteresis_campaign</code>

Execution Summary

Field	Value
Campaign name	<code>track2h_latent_state_hysteresis_campaign_2026_06_16</code>
Completed runs	6
Failed runs	0
Model type	<code>latent_state_hysteresis_probe</code>
Profiles	<code>gru , tcn</code>
Tested surfaces	<code>global , Fw , Bw</code>
Runner-level scalar first entry	<code>te_track2h_l_causal_tcn_offset_residual_global</code>
Program-level scalar winner changed	no

Directional Branch Results

Surface	Candidate	Profile	Encoder	Test MAE	Test RMSE	Val MAE
global	tcn_global	tcn	C-TCN	0.003368	0.003860	0.003543
Fw	tcn_fw	tcn	C-TCN	0.003470	0.004068	0.003565
Bw	gru_bw	gru	GRU	0.003545	0.004175	0.003837

These rows are the best scalar result per branch. The global causal-TCN entry is the campaign scalar leader, but it does not replace the forward or backward branch decisions.

Latent-State Leaderboard

Rank	Surface	Candidate	Profile	Encoder	Test MAE	Test RMSE	Val MAE
1	global	tcn_global	tcn	C-TCN	0.003368	0.003860	0.003543
2	Fw	tcn_fw	tcn	C-TCN	0.003470	0.004068	0.003565
3	Fw	gru_fw	gru	GRU	0.003537	0.004110	0.003468
4	Bw	gru_bw	gru	GRU	0.003545	0.004175	0.003837
5	global	gru_global	gru	GRU	0.003590	0.004074	0.003717
6	Bw	tcn_bw	tcn	C-TCN	0.003630	0.004312	0.003840

Encoder Split

Surface	TCN Candidate	TCN MAE	GRU Candidate	GRU MAE	Better
global	tcn_global	0.003368	gru_global	0.003590	C-TCN
Fw	tcn_fw	0.003470	gru_fw	0.003537	C-TCN
Bw	tcn_bw	0.003630	gru_bw	0.003545	GRU

The causal-TCN profile is stronger on `global` and `Fw` ; the recurrent GRU profile is stronger only on `Bw` . This suggests that short causal-history features help, but a generic hidden-state encoder is not enough by itself to solve the dispersed-offset problem.

Wave 4 series Dispersion-Aware Comparison

Surface	Wave 4.4	Wave 4.4 MAE	MDN	MDN MAE	Prob.	Prob. MAE Robust	Robust MAE	
global	tcn_g	0.003368	mdn2_g	0.003503	nll_g	0.003013	mae_g	0.003406
Fw	tcn_fw	0.003470	mdn3_fw	0.003235	nll_fw	0.003165	mae_fw	0.003146
Bw	gru_bw	0.003545	mdn2_bw	0.002658	q_bw	0.002927	sll_bw	0.003074

The latent-state campaign improves the scalar `global` surface versus the MDN and robust-loss `global` baselines, but it remains behind the Gaussian-NLL probabilistic `global` candidate. It is weaker than the existing `Fw` and `Bw` dispersion-aware leaders.

Registry Effects

The campaign runner refreshed the six family registries and updated the program registry. The program-level scalar best did not move: `te_periodic_gru_sequence_remote_Bw` remains the current scalar program winner with test MAE = 0.002344 .

Registry Scope	BestRun	Test MAE
Wave 4.4 global TCN	<code>te_track2h_l_causal_tcn_offset_residual_global</code>	0.003368
Wave 4.4 Fw TCN	<code>te_track2h_l_causal_tcn_offset_residual_fw</code>	0.003470
Wave 4.4 Bw GRU	<code>te_track2h_l_gru_offset_residual_bw</code>	0.003545
Current program scalar winner	<code>te_periodic_gru_sequence_remote_Bw</code>	0.002344

The best wave 4.4 branch is about 43.7% worse than the current program scalar winner, so it is not a direct program promotion.

TE Curve Verification Pipeline Boundary

Official TE Curve Verification Pipeline curve-first verification was not run as part of this normal campaign closeout. Under campaign governance, that remains a separate operator-approved workflow after this campaign-results report and PDF are complete.

If launched, the next TE Curve Verification Pipeline refresh should add all six wave 4.4 candidates and report global , Fw , and Bw decisions separately, comparing raw error, centered-shape error, offset behavior, amplitude, harmonic behavior, collage plots, and overlays against the completed wave 4 series , wave 5.1 , wave 3.3 , and accepted direction-parallel baselines.

Closeout Decision

wave 4.4 execution is complete: all planned candidates produced successful training artifacts, no failed run remains, registries were refreshed by the runner, and the active campaign state can be cleared.

From a modeling standpoint

- carry the causal-TCN global result forward as useful evidence that causal history helps the global scalar surface;
- do not promote wave 4.4 over the current dispersion-aware Fw or Bw leaders;
- treat GRU hidden state as branch-specific evidence, not as a generally superior hysteresis solution;
- keep latent-state / hysteresis-aware modeling available for later multi-head integration only if curve-first verification shows offset or continuity advantages that scalar MAE does not expose.

Recommended Follow-Up

1. Accept this closeout and clear the active campaign state.
2. Optionally prepare a separate operator-launched **TE Curve Verification Pipeline** verification refresh for all six **Wave 4.4** candidates.
3. If the official **TE Curve Verification Pipeline** refresh does not show a curve-quality gain, move the roadmap toward **Wave 5.2** / integrated multi-head design using **Wave 4.4** only as diagnostic evidence.